写作训练营 day 19

动物主题

为什么保护动物和植物？

Why save endangered species?

1. Medicinal
   A variety of useful medications
   Natural compounds
2. agricultural
   edible plants – wheat \ corn – depend on – world's food
   disease – resistant material
   crops – hunger problems
   diet – animal meat – protein\ nutrition – traditional survival system – main source of food
3. ecological
   foundation of healthy ecosystem
   purify the air
   clean the water
   supply them with food
   lose one plant species can trigger the loss of up to 30 other insects .
4. commercial
   commercial and recreational salmon fishing \ dolphins \ whales - jobs – annually personal income
   contributing to local and regional economies.
   Zoos – visit – pay for – lucrative industry
5. aesthetic \ recreational
   intensive tourism industry
   supply recreational, spiritual and quality – of – life values
   wildlife – related recreation – observing\ feeding \ photographing wildlife
   lodging\ equipment \ food to engage in non – consumptive wildlife recreational

外刊 1

1 **To *mimic* people means to copy their speech, dress or behaviour. In contrast, in science, *mimic* means copying ideas from nature or natural processes to prevent or solve problems or to create helpful products. The influence of this so-called *biomimicry* can be seen in a huge range of everyday products, from clothing to cars.**

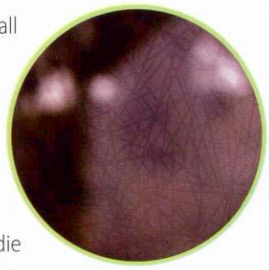
2 Perhaps the best-known example of biomimicry is Velcro®. It was invented in 1941 by a Swiss engineer called George de Mestral. One day, Mestral noticed the burdock seeds that stuck to his dog's hair. Under the microscope, he discovered that these seeds had hooks on them, so they stuck to loops on clothing or hair. Mestral copied this idea and created two strips of material: one with tiny hooks and the other with loose loops. When he put both strips together, they stuck like glue. However, unlike with glue, he could peel the strips apart and reattach them. Velcro® was initially unpopular with fashion companies, but after it was used by NASA to stop items from floating in space, it became popular with children's clothing companies. Today, it is used to fasten everything from lunch bags to shoes.

3 Another example comes from the ocean where mussels can stick to surfaces even when the water is moving very fast around them. Mussels produce a very strong adhesive[1] that works in wet **conditions**. Scientists were able to copy this material and found that it was three times stronger than other underwater adhesives. Another advantage is that it will not hurt the environment.



4 For NASA, protecting astronauts' eyes from the sun's rays is very important, but protecting their eyes from other dangerous radiation is also **essential**. Scientists studied how eagles and falcons clearly recognize their prey. Scientists discovered that the birds have yellow oil in their eyes, which filters out **harmful** radiation and allows them to see very clearly. NASA copied this oil, and it is now used by astronauts and pilots in Eagle Eyes® glasses. In addition to protecting eyes from dangerous rays, these sunglasses also improve vision in different weather conditions such as fog, sunlight or just normal light.

5 For birds, cities full of tall glass buildings are a problem. Spider's webs may be the solution. Birds cannot see the glass, so more than a billion of them fly into buildings and die every year. However, they can see spider's webs and they try to avoid them. So, engineers have used this idea in the invention of a new kind of window. These windows have tiny lines on them that look like spider's webs. Birds can see them, but the lines are so thin that humans cannot see them easily. It is likely that we will see these bird-friendly windows on more buildings in the future, and as a result, a significant drop in bird deaths.



6 As these examples **illustrate**, biomimicry appears to have an **unlimited** number of applications. It will be interesting to see which problems nature helps us solve in the future.

[1]**adhesive** (n) a material that can make things stick together

英文	汉语	默写	考频
biomimicry	n. 仿生学		
burdock	n. 牛旁属植物		
loop	n. 环		
glue	n. 胶水		
peel	v. 剥下		
fasten	v. 系紧		
Velcro	n. 魔术贴、尼龙塔扣		

the definition of biomimicry

逻辑链条

要保护植物 plants conservation---增加知识 help people advance knowledge and technology by analyzing some species -对物种的分析可以让我了解植物的进化和生存 the analysis of plants can give insight into how species evolve and survive - 进一步了解植物的特征 specific characteristics ------很多的发明都来自于对植物的观察 significant inventions and discoveries have been credited to the research on plants---------保护野生植物能够确保人类有灵感发明新东西 the preservation of all species on Earth can ensure that inventors can continue to draw inspiration from intriguing species. --- 举例论证： 牛旁属植物的种子 burdock seeds ---- 粘在衣服和毛发上 stuck on clothing and hair --- 获得启发 inspiration ---- 创造了两个带子 create two strips of material -- 粘在一起 put both strips together --- 功能： 拉开，重新粘住 peel the strips apart and reattach them --- 魔术贴的来源 Velcro --- 如今用在了儿童服装和鞋子 children's clothing and shoes \ 日常用品 daily products ---- 午餐包 lunch bags\ 宇航员 astronauts --- 防止漂浮 stop them from floating in space

每个主体段的中心思想是一样的，但是举例不一样，在一篇文章中不可以出现这么多，只需要出现一个即可

要保护动物 animal conservation---增加知识 help people advance knowledge and technology by analyzing some species---- 很多的发明都来自于对动物的观察 significant inventions and discoveries have been credited to the research on animals---------保护野生动物能够确保人类有灵感发明新东西 the preservation of all species on Earth can ensure that inventors can continue to draw inspiration from intriguing species. --- 举例论证： 贝 mussel --- 水流很快的时候也可以贴在表面 mussels can stick to surfaces even when the water is moving very fast --了解了它的特征 specific characteristics: mussels produce a very strong adhesive that works in wet conditions ---启发 stimulating --- 模仿这种物质 copy this material -- 发明了水下的粘合剂 underwater adhesives

要保护动物 animal conservation---增加知识 help people advance knowledge and technology by analyzing some species---- 很多的发明都来自于对动物的观察 significant inventions and discoveries have been credited to the research on animals---------保护野生动物能够确保人类有灵感发明新东西 the preservation of all species on Earth can ensure that inventors can continue to draw inspiration from intriguing species. --- 举例论证：科学家研究了鹰 scientists studied eagles --- 了解到鹰的视力很好 noted their clear vision. ---- 发现了有黄色的眼油 Discovery: birds have yellow eye oil,-- 过滤有害的辐射 filters harmful radiation -- 提升视力 enhances visual clarity. --- 应用于 NASA: copied this natural filter --- 创造了鹰眼科技 created Eagle Eyes technology. ------ 应用场景 Application: used by astronauts/pilots, protects eyes from dangerous rays, improves sight in fog/sunlight/normal light.

要保护动物 animal conservation---增加知识 help people advance knowledge and technology by analyzing some species---- 很多的发明都来自于对动物的观察 significant inventions and discoveries have been credited to the research on animals---------保护野生动物能够确保人类有灵感发明新东西 the preservation of all species on Earth can ensure that inventors can continue to

draw inspiration from intriguing species. --- 举例论证： prevent birds flying into Buildings---spider's wed ---can be seen by birds---new kind of window

to protect birds from the buildings with windows， which cause many birds to die ——copy---birds can detect spiders web ——a new type of windows be invented with tiny lines——to avoid bird

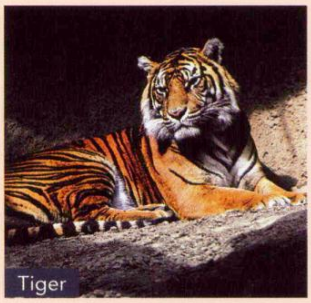
birds--cannot see glasses---fly into/ die vs. see spider's web - try to avoid--- engineers: new window---birds can see - avoid death

外刊 2:

1 An **endangered species** is a group of animals or plants which could soon become extinct. Extinction happens when the last animal of the species has died out and there will be no more. Many species are nearly extinct and could disappear from the Earth very soon if we don't do anything to save them. There are many reasons why species become endangered, but most harm to species is **due to** human activities such as habitat destruction, hunting and overfishing.

2 Habitat destruction is the main reason why animals become endangered. This happens in two ways. First, when humans move into a new area, they cut down trees to build houses and farms. This **destroys** the animals' habitat – the **natural** environment where plants or animals usually live – and leaves them without food. Animal habitats are also destroyed because of pollution. Dirty water from factories, which contains **chemicals**, ends up in rivers, and poisons used on farmland may even kill animals which live in the area.

3 Endangered species are also the result of hunting and fishing. Animals such as the Arabian oryx are nearly extinct because of the high price of their meat. Other animals are killed for their fur, bones or skin – or just for sport. For example, some seal species are now almost extinct because they are killed for their fur to make coats. Tigers are shot to make medicine and tea from their bones, and crocodiles are caught to make bags and shoes. Large sea creatures like whales, tuna and sharks have all become endangered species because of overfishing – too many are caught to make special dishes that people like to eat, such as shark's fin soup or sushi.


Tiger

4 What steps can individuals and governments take to **protect** more animal and plant species from becoming endangered? We should try not to **pollute** natural areas, and farmers or companies who destroy animal habitats should face a financial penalty. The public can help out by refusing to buy products made from animals' body parts, such as seal fur coats or crocodile bags. Governments can help, too, by making it against the law to hunt, fish or trade in endangered species. They can also provide funding for animal sanctuaries and zoos. These protect animals from extinction by breeding more endangered animals, which can later be released into the wild. If we all cooperate by taking these steps, we will protect our planet so that our children and their children can enjoy it, too.

steps:
try not to pollute natural areas
financial penalty -- farmers\ companies

refuse to buy products made from animals' body parts
make it against the law to hunt \ fish
provide funds
zoo
breed

外刊 3

Student A

Invasive species are plants and animals which arrive in an area where they are not native, usually due to human activity. For example, a species of shellfish might attach itself to the outside of a ship travelling between countries and enter a new environment in this way. Invasive species are often able to grow quickly in their new homes because they have no natural enemies. As a result, they may replace or damage native plants and animals which live in the same environment. One example is the case of grey and red squirrels in the UK.

Student B

Red squirrels used to be a **common** sight in British forests and countryside. Then, in the 1870s, the grey squirrel was **introduced** from North America because rich people thought the squirrels looked fashionable in the gardens of their large homes. Today, only about 140,000 red squirrels remain, mostly in Scotland. In contrast, grey squirrels are now extremely common and seen as **major** pests due to the damage they cause to plants and houses. While red squirrels are protected, grey squirrels can be legally trapped and destroyed.

Student C

Three more differences explain why red squirrels have lost out in the competition with grey squirrels. First, red squirrels live high up in the trees, whereas grey squirrels spend most of their time on the ground. This means that any loss of forest habitat greatly affects the red squirrel population. Another reason is that grey squirrels are more intelligent and can adapt to new situations more easily than red squirrels. For example, they can survive in an urban environment because of their ability to use food provided by humans. A third problem for the red squirrel is **disease**. Both squirrels carry parapoxvirus. The virus does not seem to affect grey squirrels, but it is fatal to reds.

Student D

In conclusion, there does not seem to be much that scientists can do to help red squirrels **survive** in Britain. Some politicians support destroying populations of grey squirrels, but many British people would contend that this is **cruel**. Red squirrels have been successfully reintroduced from other countries and they could be protected in places where there are no grey squirrels, such as the Isle of Wight. However, some people question whether Britain should protect red squirrels at all. Worldwide, they are not an endangered species. Considering the evidence, saving the red squirrel may be a waste of British government money. Government conservation funding should instead be spent on other endangered animals.

外刊 4: 英国卫报

Reviving the woolly mammoth isn't just unethical. It's impossible

At a time when US scientists are under attack from their own government, the illiteracy around these elephantine fantasies is dangerous



You will never ever see a living woolly mammoth. While this is an obvious truth to most geneticists, zoologists and mammoth experts, the endless promises that you might get to meet an extant version of this very-much extinct elephantid apparently necessitate me typing it.

The latest on the conveyor belt of mammoth resurrection stories came this week in the form of a slightly hairy mouse. Colossal Biosciences, the US company behind the "woolly mouse"

and ensuing media frenzy, published a non-peer-reviewed paper in which it has genetically engineered a mouse to express a gene that relates to mammoth hair, resulting in a mouse with slightly longer hair than normal.

Mice are a crucial part of the story of genetics, not least because of their utility in the study of human genes, and particularly disease. But this unusually hirsute mouse was not created for any such noble purpose but rather as a farcical little sideshow for the fantasy that one day Colossal will breed a living mammoth.

"The woolly mammoth is a vital defender of the earth," it claims on its website, despite the fact that mammoths are extinct and therefore – I feel this shouldn't really need saying – have literally no role in defending the Earth. Colossal states that packs of mammoths were a crucial part of tundra ecosystem management in their heyday tens of thousands of years ago – that much is true – and that bringing them back might be a useful tool in the fight against global warming. It is a ridiculous suggestion for tackling the climate crisis – and more than that, it is scientific folly.

How is it folly? Let us count the ways. A few years ago, the first mammoth genome was sequenced from remains acquired off eBay. You can get some today from as little as a tenner. The genome sequence allowed scientists to piece together some aspects of the lives and the biology of these magnificent beasts, as indeed we have for many animals – including humans – that are long dead. This is good science, gifting us insights into the adaptations that helped mammoths evolve to suit their environments. What the ancient genome of the sadly departed does not allow us to do is bring them back.

DNA from dead creatures is profoundly degraded, chewed up by time and bacteria, and what we actually recover is fragments. When scientists unveiled the "complete" genome of our Neanderthal ancestors in 2009, it was in fact sparse bits and pieces of the interesting parts, the genes, which we could usefully compare with our living DNA. But genes make up a tiny proportion of the total amount of DNA in a living creature, most of which is lost in time, like

tears in rain. Much of that lost DNA has function in living cells and is species specific. This is not cloning as per the famous Dolly, whose actual complete genome was taken from a living sheep. This is not photocopying – it's trying to assemble a book with only a clutch of sentences.

Colossal knows this and is proposing to modify the genes of an Asian elephant so they are more like mammoth genes. The proposed resulting creature would be a mixture of elephant and mammoth. These two elephantids parted ways on their evolutionary trees roughly 6m years ago. That was at about the same time we separated from what ended up as chimpanzees. That's how different they are; that's the equivalent hybrid being proposed.

Cast your moral distaste over that particular monster aside and just consider that we don't know the gestation period for a mammoth. We don't know how big a mammoth neonate would be. We don't know if a mammoth could digest Asian elephant milk. We would be performing artificial insemination and experimental IVF on an elephant that I am guessing did not give informed consent for this procedure.

Should Colossal manage to overcome all of these apparently insurmountable scientific barriers, it will have an Asian elephant mother giving birth to a different species into a social group with which it has no affiliation in an environment it has not evolved to be part of because it ceased to exist many millennia ago.

And it will be utterly alone. The best possible outcome will be one single boutique animal that is profoundly confused. More likely it will die very quickly. At present the Pyrenean ibex is the only animal brought back from extinction, via cells taken from the last known member of its wild goat species. Born to a surrogate in 2003, the kid immediately died, making it the only species to have gone extinct twice. The mammoth, should Colossal succeed, would surely be the second.

The absurd and frankly ghoulish claims about the mammoth's resurrection amount to a textbook case of science miscommunication and hubris. At a time when US scientists are

under attack from their own government, the illiteracy around these elephantine fantasies is not just vexing but dangerous. The Trump administration's threatened cuts span all scientific disciplines, but most pertinently to conservation and climate-crisis research. We are witnessing – and party to – the greatest biodiversity and species loss in human history. More than ever, science needs money, public support, and government backing. Perhaps focusing our efforts on preserving the millions of threatened creatures that actually exist should be the priority in these hostile times.

The only way you will ever see a living mammoth is if our physicist friends finally crack time travel. I am a mere geneticist, but my understanding is that this remains very much in the realm of fiction. Perhaps in the meantime we could direct our scientific excitement and energies towards real problems, things on which millions of lives depend, rather than on this mammoth circus of macabre fantasy and moral bankruptcy. ■

保护猛犸象 protecting mammoth -- 猛犸象是地球的关键保护者 mammoth is a vital defender of the earth ---猛犸象群 苔原带生态系统的关键部分 the packs of mammoths were a crucial part of tundra ecosystem management in their heyday tens of thousands of years ago -- 复活猛犸象 bring them back --- 一个有效的工具去对抗全球变暖 might be a useful tool in the fight against global warming

反驳上面的说法：这是很荒唐的 it is a ridiculous suggestion for tackling the climate crisis -- 从科学的角度上是荒谬的 it is scientific folly -- 原因： 猛犸象的基因库 mammoth genome was sequenced from remains acquired off eBay --拼凑到一起 piece together -- 理论上可以复原猛犸象 some aspects of the lives and the biology of these magnificent beasts --- 有前途 promise -- 但是基因库不全 incomplete ancient genome --- DNA 不足 DNA from dead creatures is profoundly degraded -- 随着时间和细菌不断磨损 chewed up by time and bacteria -- 复原的都是碎片 what we actually recover is fragments -- 无法复原 not allow scientists to bring them back

不复原已经灭绝的动物-- 猛犸象复活 - 遇到的难题很多 - 不知道怀孕期多久 do not know the gestation period for a mammoth --- 不知道一个新生儿有多大 do not know how big a mammoth neonate would be ---不知道他吃喝什么 do not know if a mammoth could digest Asian elephant milk --- 不适应环境 has no affiliation in an environment it has not evolved to be a part of

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猛犸象复活 mammoth's resurrection -- 需要很多的钱 science needs money, public support, and government backing --- 现在的时期来看， 正处在一个前所未有的生物多样性的丢失上 the greatest biodiversity and species loss in human history --- 财政上预算不足 deficit \ not

enough budget --- 砍了很多的科学科目的研究周期 cut span all scientific disciplines -- 把大部分的钱聚焦到了直接相关的项目上 but most pertinently to conservation and climate- crisis research - 精力应该放在保护现存的濒临灭绝的动物上 focusing our efforts on preserving the millions of threatened creatures that actually exist should be the priority in these hostile times----关注点、兴奋点都要在具体的问题上 direct our scientific excitement and energies towards real problems, things on which millions of lives depend ------无暇顾及猛犸象的复原 rather than on this mammoth -- 应该在科幻领域 in the realm of fiction

动物实验

支持动物实验

1. 科技结果和动物实验结果相差较大 no current new approach methodologies (NAMs)gets anything close to the real biology. ----- 机体很复杂， 现实很难 biology is extremely complicated and the reality may prove much harder. -------- 各种内部和外部原因会影响 With complex areas of biology, such as the brain and behaviour, reproductive and endocrine systems, the immune system, tumour biology, or where there is a need to account for ageing, altered physiologies, environmental effects, etc ---很难想象如果没有动物实验，只有科技模拟之下的反映 It is difficult to envisage that understanding the biology of complex disease such as those involving developmental changes, ageing or requiring understanding of the interactions between biology and the environment will be replaced by these technologies 这个科技的方式是不可能帮助作出新的发现和治疗疾病的 this strategy is unlikely to help new discoveries relevant to treating or preventing disorders and diseases in humans or animals. --- 真实性和合理性��质疑 validation - 安全问题 safety

2. 目前市面上的新方法都是基于动物制品的 using new approach methodologies (NAMs) still rely on products obtained from animals. --- 举例论证： 细胞的成长和生存离不开动物 While the cells themselves can be derived from human stem cells, the matrix 条件 on which they grow, and the growing mediums needed for them to develop and survive, require animal-derived tissue. -- 血清为例，对于动物实验必不可少， 没有合成的东西可以替代血清 culturing in products such as foetal= fetal 知名的 bovine 牛科类动物 serum 血清, which contain essential growth factors and which synthetic alternatives cannot currently replace.

3. 动物实验很靠谱 it is reliable, ---- 提供模型、输出信息 that animals provide sufficiently good models of human biology and diseases to yield relevant information, and that, ---------可以预测在人身上的反映 predictive value for human outcomes---- 有利于人的身体健康 consequently, its use provides major human health benefits.-- 准确度得以保证： validity of animal experimentation

4. 人和动物很相似 considerable congruence were shown between an animal model and its

corresponding human disease -- 人类和大猩猩以及老鼠的基因高度相似 we share more than 96% of our genes with chimpanzees and at least 85% with mice ---- 我们可以从动物实验中获取信息 some information obtained from animal experiments ----动物实验的成功对于药物创新和突破很关键 animal testing has been crucial for biomedical innovation and success

5. 反驳伦理到的时候反驳段
伦理困境： ethical dilemma ----- animal suffering vs human suffering -- animal--

6. 动物好控制 control – 问题 – 杀害、 控制在一定的范围内 人： 不安全 – 流动 – 风险扩散 the expansion

反对动物实验

1. 可以使用科技减少对动物实验的依赖 using new technologies to reduce the reliance on animal experiments -- 科技的进步 - 电脑分析是产生大量的细节的信息，比如说去观测细胞的基因活动 Advances in the ability to generate large amounts of detailed information about gene activity in cells and computer analysis have been amazing ----- 各种新方法可以替代动物实验 growing miniature human organs in the lab and modelling human biology with AI, are emerging to replace procedures previously carried out on animals. 废除动物实验 phase out animal testing

2. 动物实验的不可靠性和局限性已经呈现的很多了 unreliability and limitations of animal experimentation have increasingly been acknowledged -- 解释说明：实验室的操作步骤和环境会对实验结果产生影响 Laboratory procedures and conditions exert influences on animals' physiology and behaviors that are difficult to control and that can ultimately impact research outcomes -不通风、噪音 captivity\ artificial lighting, human-produced noises, and restricted housing environments -- 都会影响实验结果 causing distress and abnormal behaviors among animals （实验室里面虐待动物的现象 - 伦理上）

3. 学术化的写法
人和动物的区别大于相似性 the differences among species along these subtler biological dimensions far outweigh the similarities, ----------具体阐述有哪些不一样： interspecies differences in physiology, behavior, pharmacokinetics, and genetics would significantly limit the reliability of animal studies--------- 跨物种的不同，影响药物效果 The lack of sufficient congruence between animal models and human diseases is another significant obstacle to translational reliability -- 人的疾病被人造到动物身上， 限制作用明显 Human diseases are typically artificially induced in animals, but the enormous difficulty of reproducing anything approaching the complexity of human diseases in animal models limits their usefulness ---------- 动物身上有效果， 但是人身上会失效 Even if the design and conduct of an animal experiment are sound and standardized, the translation of its results to the clinic may fail because of disparities between the animal experimental model and the human condition ---- 临床实验失败

human clinical trials fail 要素 poor methodology, publication bias, lack of preexisting disease and medications, wrong gender or age
稍微不那么正式一点
人和动物的区别大于相似性 the differences among species along these subtler biological dimensions far outweigh the similarities ----- 具体的不一样的方面： Humans and animals can differ significantly from each other, for instance in terms of anatomy, organ function, metabolism and nutrition - 人有个体性 individual -- 各个要素都会影响药效 Age, gender, background or environment also play a large part in how one reacts to certain substances ------ 举例： women \ men, adult \ child 用的药物都不同 -- 人和人之间不兼容，人和动物为什么会兼容 if humans are not comparable amongst themselves, how should animal results be transferred to humans? ---- 很多的药物在动物神圣有效但是在人身上无效 there are many substances that are effective and safe in animals and ineffective in humans and/or cause strong side effects

4. 对于动物的生命的尊重 Respect for the lives of animals is the main reason many people reject animal experiments. -----伦理的合理性 question the ethical justifiability of animal experiments -- 人和动物是平等的 reciprocal fairness between humans and animals--- 动物实验会让动物苦难，甚至死亡 The extent of animal suffering or the possible death of the animals during the experiment ---- 动物实验的终点是动物的死亡 animal experiments, in general, often end in the death of the animals

5. 驳斥： 动物实验有利于研发药物--作用于动物身上， 挽救动物的生命 --- 反驳他： 动物实验不会治愈动物的疾病， 很多的疾病动物不会得 Many diseases that occur in humans do not occur or are rare in animals. For example, animals do not get Alzheimer's or Parkinson's diseases. However, in order to be able to research them, the animals are genetically, surgically, medically or behaviourally manipulated so that they develop symptoms similar to those of the disease. Common "animal models" for cancer are, for example, "generated" by genetic manipulation or the injection of human cancer cells into mice. A tremor reminiscent of Parkinson's disease is also displayed by monkeys, mice, or rats who have a specific neurotoxin injected into the brain. In depression research, rats are placed in a water tank from which they cannot escape. If they stop swimming, they are considered to be depressive -- 除了给动物带来痛苦什么都没有带来 （也是可以用来阐述动物实验不人道）

6. 很多动物实验没有必要 unnecessary --- 化妆品领域 cosmetic research ---为了人类自身的利益去滥用动物违背了伦理 it is against moral principles to exploit living animals for humans' self – interest –

Zoos – 意义：补充材料 剑桥 14 test 4 passage2

In my view, it is perfectly possible for many species of animals living in zoos or wildlife parks to have a quality of life as high as, or higher than, in the wild. Animals in good zoos get a varied and high-quality diet with all the supplements required, and any illnesses they might have will be treated. Their movement might be somewhat restricted, but they have a safe environment in which to live, and they are spared bullying and social ostracism by others of their kind. They do not suffer from the threat or stress of predators, or the irritation and pain of parasites or injuries. The average captive animal will have a greater life expectancy compared with its wild counterpart, and will not die of drought, of starvation or in the jaws of a predator. A lot of very nasty things happen to truly 'wild' animals that simply don't happen in good zoos, and to view a life that is 'free' as one that is automatically 'good' is, I think, an error. Furthermore, zoos serve several key purposes.

Topic sentence:

It is perfectly possible for many species of animals living in zoos or wildlife parks to have a quality of life higher than in the wild.

Develop：comparison

Zoos 1. Get a varied and high- quality diet with all the supplements required. 2. Any illness will be treated 3. Have a safe environment – they are spared bullying and social ostracism by others of their kind ---- Captive animal will have a greater life expectancy

Wild: 1. Die of starvation and drought. 2. Suffer from the irritation and pain of parasites or injuries. 3. Suffer from the threat or stress of predators\ die in the jaws of a predator 4. A lot of nasty things happen to truly wild animals – cannot guarantee the life span

Firstly, zoos aid conservation. Colossal numbers of species are becoming extinct across the world, and many more are increasingly threatened and therefore risk extinction. Moreover, some of these collapses have been sudden, dramatic and unexpected, or were simply discovered very late in the day. A species protected in captivity can be bred up to provide a reservoir population against a population crash or extinction in the wild. A good number of species only exist in captivity, with many of these living in zoos. Still more only exist in the wild because they have been reintroduced from zoos, or have wild populations that have been boosted by captive bred animals. Without these efforts there would be fewer species alive today. Although reintroduction successes are few and far between, the numbers are increasing, and the very fact that species have been saved or reintroduced as a result of captive breeding proves the value of such initiatives.

TS: zoos aid conservation.

Zoos – reservoir – breed up – against the population crash and extinction – rewilding – wild population will be boosted – save species

Zoos also provide education. Many children and adults, especially those in cities, will never see a wild animal beyond a fox or pigeon. While it is true that television documentaries are becoming ever more detailed and impressive, and many natural history specimens are on display in museums, there really is nothing to compare with seeing a living creature in the flesh, hearing it, smelling it, watching what it does and having the time to absorb details. That alone will bring a greater understanding and perspective to many, and hopefully give them a greater appreciation for wildlife, conservation efforts and how they can contribute.

T.S. zoos also provide education.

Television – documentaries -- detailed and impressive

In reality: seeing a living creature in the flesh – hearing\ smelling \ watching – have the time to absorb details --- understanding and perspective – appreciation for wildlife

In addition to this, there is also the education that can take place in zoos through signs, talks and presentations which directly communicate information to visitors about the animals they are seeing and their place in the world. This was an area where zoos used to be lacking, but they are now increasingly sophisticated in their communication and outreach work. Many zoos also work directly to educate conservation workers in other countries, or send their animal keepers abroad to contribute their knowledge and skills to those working in zoos and reserves, thereby helping to improve conditions and reintroductions all over the world.

Zoos 改成 museums wildlife – artefact

Why not museums?

Zoos also play a key role in research. If we are to save wild species and restore and repair ecosystems we need to know about how key species live, act and react. Being able to undertake research on animals in zoos where there is less risk and fewer variables means real changes can be effected on wild populations. Finding out about, for example, the oestrus cycle of an animal or its breeding rate helps us manage wild populations. Procedures such as capturing and moving at-risk or dangerous individuals are bolstered by knowledge gained in zoos about doses for anaesthetics, and by experience in handling and transporting animals. This can make a real difference to conservation efforts and to the reduction of human–animal conflicts, and can provide a knowledge base for helping with the increasing threats of habitat destruction and other problems.

Research 举例论证 – 并列结构 – be less risk and fewer variables – real changes can be effected on wild population

Oestrus cycle of an animal \ breeding rate – manage wild populations

Habitat destruction \ reduce the conflict between humans and animals

Transporting and handling animals – knowledge gained in zoos about doses for anesthetics

Rewilding – restore and repair ecosystem – related documentary to help you know how the species live, act and react.

关在动物园的坏处：
Morally wrong to keep animals in zoos --- this is against their instincts – animals are captured and kept in the cages – in their traditional home areas, they can roam freely and maintain their natural behavioral traits: interacting with other members of a same species - animals lose freedom to hunt for food – they are either overfed or starving

Some zoos train animals for the purpose of entertaining tourists --- animals are beaten or whipped if failing to behave as instructed by trainers – animal welfare campaigners argue that we have no right to use animals for entertainment and profit.

举一反三： lab animals\ circus – subject to maltreatment – poor living condition

Zoos - museum – library 用一样的素材， 改部分的内容即可

相关的题目：
动物保护
There are many extinct animal species in the world nowadays. Some people say we should protect these animals from dying out, while others say we should concentrate more on problems of human beings. Discuss both views and give your opinion.

Many species of plants and animals are declining in the world. Give possible reasons and offer solutions.

Human activities have negative effects on plants and animal species. Some people think it is too late to do anything about this problem. Others believe that effective measures can be taken to improve this situation. Discuss both views and give your opinion.

动物实验
Nowadays animal experiments are widely used to develop new machines and to test the safety of other products. Some people argue that these experiments should be banned because it is morally wrong to cause animals to suffer, while others are in favor of them because of their benefits to humanity. Discuss both views and give your own opinion.

动物园的功能
Some people argue that the purpose of zoos is only to entertain people. What do you think ?What other purposes of zoos?

Some people believe that zoos serve useful functions for education and conservation. Others believe that it is cruel to keep animals in captivity. Discuss both views and give your opinion